TEXORA · LECTURE SHEET

# Machinery Used for Dyeing Different Textile Materials

*Winch, jigger, jet, overflow, package and pad ranges — what moves, at what liquor ratio and temperature, decides what can be dyed and at what cost.*

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 6 · LEVEL 3 TERM 1

## 1. What the machine must do

Every dyeing machine solves the same transport problem: bring fresh dye liquor into contact with every part of the goods, uniformly, at a controlled temperature programme, for the process time — then wash. Machines differ in **what moves** (liquor, goods, or both), in the **form of goods** (fibre, yarn package, piece, garment), and in attainable liquor ratio and temperature. Choosing wrongly shows up as unlevelness, creasing, package channeling or fibre damage that no recipe can rescue.

## 2. Loose-fibre and yarn-package machines

**Loose stock dyeing:** fibre packed in a cage; liquor pumped through it (radial flow). Used for melange yarns and wool tops. **Package dyeing:** yarn wound on perforated cones/cheeses stacked on spindles in a pressurised vessel; liquor flows inside-out then outside-in in alternating cycles. Critical variables: package density uniformity (soft winding for dyeing), flow reversal, and air removal. LR ≈ 1:8–1:12; HTHP versions dye polyester yarn at 130 °C. Hank dyeing (reel machines) persists for bulky wool/acrylic yarns where package pressure would crush loft.

## 3. Winch (beck) — goods move through a stationary bath

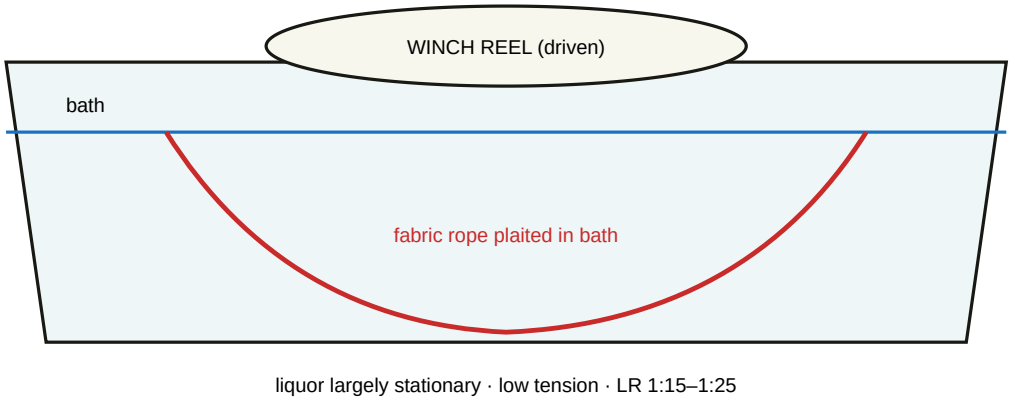



Fig. 1 — Winch dyeing machine: the reel lifts the fabric rope and lets it plait down into the bath.

The winch lifts a rope of fabric over a driven reel and lets it slide back into the bath, where it lies plaited for most of the cycle. Tension is minimal — kind to knits, wool and delicate goods — but the rope spends long periods folded in the same creases (rope marks) and the bath is large (LR 1:15–1:25), so salt and energy costs are high. Temperature is limited to the boil in open winches.

## 4. Jigger — goods travel between two rollers through a shallow bath

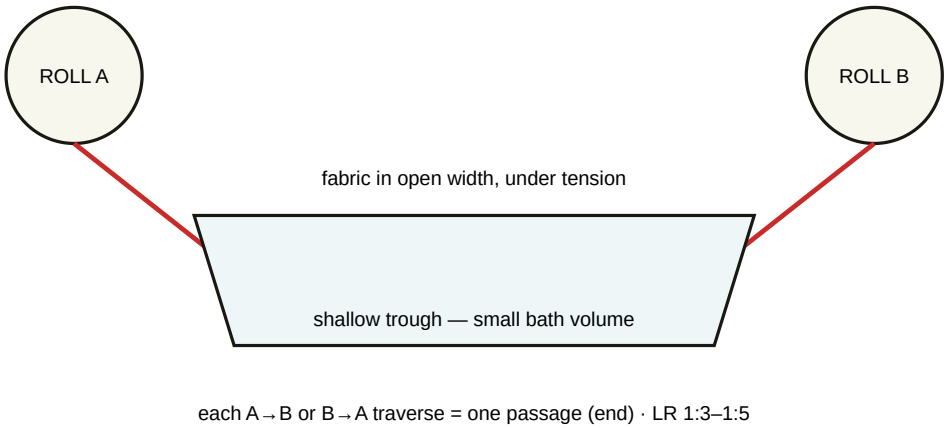



Fig. 2 — Jigger: fabric winds from roll to roll through a shallow trough, in open width.

Fabric in **open width** winds from one main roller to the other through a shallow trough; each traverse is a *passage* or *end*. Advantages: no rope creases, very low LR (1:3–1:5), ideal for woven cotton with reactive, vat and direct dyes. Risks: **ending** (shade drift end-to-end as the bath exhausts between passages) and **listing** (side-to-centre variation), plus tension unsuited to stretchy knits. Pressurised HT jiggers reach 130 °C for polyester blends.

## 5. Jet and overflow — both goods and liquor move

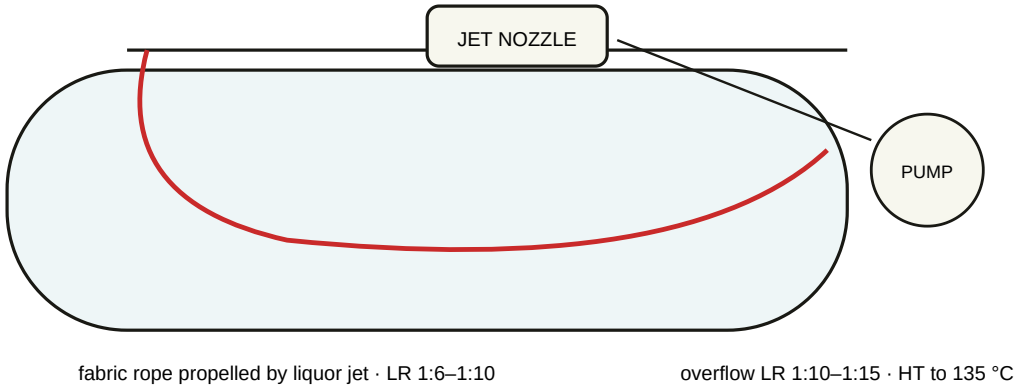



Fig. 3 — Jet dyeing machine (schematic): a liquor jet transports the fabric rope around a closed loop.

In a **jet** machine a venturi nozzle uses pumped liquor to propel the fabric rope around a closed circuit at high speed — vigorous interchange, excellent levelness, HTHP to 130–135 °C, LR 1:6–1:10; the workhorse for polyester and knits. **Overflow** machines use a gentler flooded transport (fabric carried by overflowing liquor, low nozzle pressure) for crease-sensitive and delicate goods. **Airflow** machines move the rope with air and a minimal liquor film (LR as low as 1:3–1:5), cutting water and chemicals sharply. Beam dyeing (fabric wound on a perforated beam, liquor pumped through) suits open-width synthetics without any rope handling.

## 6. Continuous ranges and garment dyeing

**Pad–dry–cure/steam** ranges colour woven goods continuously for reactive and pigment systems (Topic 4 flow). **Garment dyeing** uses rotating perforated drums (paddle/extractor-type machines) so made-up garments dye and wash in one vessel — the basis of pigment garment effects and reactive garment dyeing.

## 7. Selection table

| MACHINE | LR | MAX TEMP | BEST FOR |
|---|---|---|---|
| Winch | 1:15–1:25 | 98 °C | Delicate knits, wool, low tension |
| Jigger | 1:3–1:5 | 98 °C (HT 130 °C) | Open-width woven cotton |
| Jet (HTHP) | 1:6–1:10 | 135 °C | Polyester, knits, rapid level dyeing |
| Overflow | 1:10–1:15 | 98–130 °C | Crease-sensitive, delicate synthetics |
| Package | 1:8–1:12 | 135 °C | Yarn on cones/cheeses |

| | | | |
|---|---|---|---|
| Pad range | — (pick-up) | Cure 150 °C | Continuous woven production |

## 8. Glossary — key terms

- **Passage (end):** one traverse of fabric through a jigger trough.
- **Ending / listing:** end-to-end / side-to-centre shade variation.
- **HTHP:** high-temperature high-pressure dyeing above 100 °C.
- **Channeling:** liquor taking preferred paths through a package, causing patchiness.

## 9. Key points

- Classify machines by what moves and by goods form; LR and temperature ceiling decide running cost and dye class.
- Winch = gentle but thirsty; jigger = frugal open-width; jet = fast, hot, versatile.

## 10. Summary

Machine choice is applied transport physics: the winch's patience, the jigger's economy, the jet's vigour and the package machine's precision each fit a goods form and a dye system. A correct recipe on the wrong machine still fails; the selection table is the first page of every production plan.

## 11. Practice

1. Choose machines, with reasons, for: (a) 500 kg polyester knitwear, disperse dyes; (b) woven cotton poplin, reactive dyes, crease marks unacceptable; (c) acrylic yarn for melange knitting.
2. A jigger runs LR 1:4 on 200 kg fabric while a winch needs 1:20 for the same goods. Calculate both bath volumes and the salt needed at 50 g/L in each.
3. Why does package dyeing alternate flow direction, and what defect appears if winding density varies?

### Answer key

1. (a) HTHP jet — 130 °C disperse dyeing with vigorous circulation. (b) Jigger — open width, no rope creases, low LR. (c) Package (or hank) dyeing of the yarn before knitting — melange requires dyed yarn, not dyed fabric.
2. Jigger bath = 200 × 4 = 800 L → salt 50 × 800 = 40 kg. Winch bath = 200 × 20 = 4,000 L → salt 200 kg. Five times the salt for the same shade — the economic core of machine selection.

3. Alternation evens out flow resistance differences so inside and outside of the package dye equally. Uneven winding density channels liquor through loose zones; tight zones are starved and dye pale and unlevel.

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